

Algorithmic Supervision of High-Frequency Financial Flows

A Permissioned Blockchain and Agent-Based Simulation for AML Compliance in Nepal

Dikshant Pandey

Kathmandu, Nepal

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Version 2.0: 100k transaction reproducible simulation

Abstract

This research presents the Nepal Compliance Ledger (NCL), a regulatory technology prototype for supervising high-frequency financial flows before settlement. The updated implementation combines an agent-based transaction market, sliding-window velocity checks, bounded graph-cycle detection, and a permissioned blockchain audit layer with Merkle roots, hash-linked blocks, policy-version binding, and validator quorum signatures. A deterministic simulation with 10,000 agents and 100,000 transactions produced 38,566 true positives, 39 false positives, 6,601 false negatives, and 54,794 true negatives. The system achieved 99.90% precision, 85.39% recall, and an F1 score of 0.9207 while sealing 196 auditable blocks with intact chain verification. These results suggest that a hub-and-spoke compliance ledger can reduce AML latency while preserving a tamper-evident regulatory audit trail.

1 Introduction

Anti-money laundering (AML) enforcement in developing financial systems faces a timing problem. Suspicious flows are often detected after settlement, after funds have already moved across institutions or been fragmented into low-value transfers. Structuring, also known as smurfing, exploits this weakness by splitting a larger illicit amount into many smaller transactions that sit below scalar reporting thresholds.

NCL reframes the problem as one of transaction velocity and graph topology. Instead of asking only whether a single payment exceeds a threshold, the system evaluates whether a sender is moving too frequently inside a rolling time window and whether a new edge closes a short circular path. The updated implementation also records every compliance decision into a permissioned blockchain-style ledger so that banks and regulators can audit a shared, tamper-evident history.

2 Methodology

2.1 Hub-and-Spoke Architecture

The project preserves the original hub-and-spoke design: commercial nodes submit transaction requests, a regulatory compliance brain evaluates them, clean transactions proceed to participating banks, and every decision is written into an immutable audit ledger with a regulator dashboard feed.

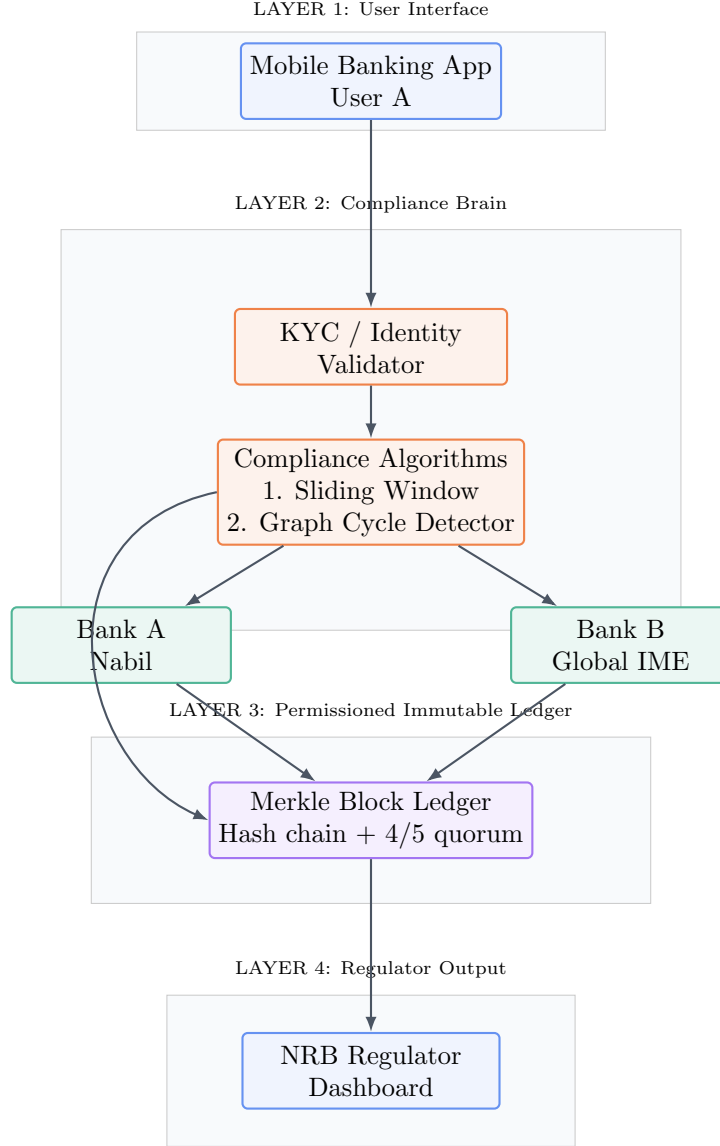


Figure 1: NCL hub-and-spoke architecture, preserving the original system design while replacing the append-only log with a permissioned block ledger.

2.2 Sliding-Window Velocity Protocol

For each sender a , the engine stores a rolling log of recent transaction times. A transaction at time T is blocked when the number of transactions in the last W seconds exceeds a policy threshold τ :

$$\text{VelocityCount}(a, T) = \sum_i \mathbf{1}(T - t_i < W)$$

In the implementation, $W = 300$ seconds and $\tau = 5$. A deque-backed log is used so outdated timestamps are pruned in amortized constant time.

2.3 Graph Cycle Detection

Layering is modeled as a short cycle in the directed transaction graph. For a proposed edge (u, v) , NCL checks whether a path already exists from v back to u with depth at most k :

$$v \rightarrow \dots \rightarrow u, \quad |P| \leq k$$

The implementation uses bounded DFS with $k = 3$ and a 900-second graph memory window. This keeps the detector focused on recent circular flows rather than every historical transfer.

2.4 Permissioned Blockchain Audit Layer

The advanced implementation upgrades the earlier append-only ledger into a permissioned block ledger. Each transaction decision is converted into a canonical audit record and batched into blocks of 512 records. A block header contains:

- previous block hash,
- Merkle root over transaction decision records,
- policy-version hash,
- proposer node,
- simulated validator signatures from a 4-of-5 quorum.

The chain verifier recomputes every Merkle root, block hash, previous-hash link, and quorum signature. A run is accepted only when the complete audit chain verifies as intact.

3 Experimental Setup

Parameter	Value
Agents	10,000 autonomous actors
Malicious population	5% of agents
Transaction volume	100,000 transactions
Transaction mix	55% honest, 30% structuring, 15% ring-layering attempts
Sliding window	300 seconds
Velocity threshold	5 sender transactions per window
Cycle detector	Depth-3 DFS with 900-second graph memory
Ledger	512-record blocks, SHA-256 hash chain, Merkle roots, 4-of-5 validator quorum

4 Results

100,000	99.90%	85.39%	0.9207
Transactions	Precision	Recall	F1 Score
deterministic ABM run	39 false positives	6,601 false negatives	balanced detection metric

Confusion Matrix	Count	Ledger / Runtime	Value
True positives	38,566	Sealed blocks	196
False positives	39	Block size	512
False negatives	6,601	Validator quorum	4/5
True negatives	54,794	Processing throughput	102,948 tx/s

Detection Breakdown	Count
Velocity flags	38,271
Cycle flags	334
Approved transactions	61,395
Ledger chain integrity	Intact

5 Discussion

The upgraded simulation shows that most malicious activity is captured by velocity detection, while graph-cycle detection adds coverage for circular layering behavior. Precision remains high because honest users rarely exceed the rolling velocity threshold, and the bounded graph memory reduces accidental long-horizon cycle matches.

The ledger layer is not included for speed alone. Its purpose is institutional accountability. A centralized database can store the same rows, but a permissioned ledger creates a shared audit object: each block binds transaction decisions to a policy hash, previous block hash, Merkle root, and validator quorum. This makes later alteration detectable without requiring every participant to trust a single administrator.

6 Limitations and Future Work

- The validator signatures are deterministic simulation signatures, not production cryptographic keys managed by real bank HSMs.
- The ABM uses synthetic behavior distributions; deployment would require calibration against legally shareable banking telemetry.
- The current graph detector uses bounded DFS; larger deployments should evaluate streaming graph anomaly methods.
- The policy engine is rules-first; a future version can add supervised risk scoring while keeping rule decisions explainable.

7 Conclusion

NCL demonstrates how real-time AML supervision can be represented as a pre-settlement compliance problem. The advanced implementation scales the simulation to 100,000 transactions and adds a permissioned blockchain audit layer with Merkle verification and validator quorum. The result is a research prototype that connects transaction monitoring, graph-based detection, and tamper-evident regulatory auditability in a single reproducible system.

References

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